Scorecard Update

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Overview

This markdown shows the data logging process of each round, and shows performance trends and metric averages

Set-up Environment

Attach Packages

```r
library(golf)
library(tidyverse)
library(lme4)
library(mgcv)
library(brms)
library(DBI)
library(duckdb)
# library(blastula)
# library(keyring)
# library(mailR)
library(emayili)
```

Record New Scorecard

Input the Scores Data

```r
# required inputs to fill out the scorecard

round_course <- 'Randolph North'
round_date <- '2026-04-19'
round_tees <- 'blue'

hole_scores <- c(5, 5, 6, 4, 5, 4, 4, 4, 8,
                 4, 3, 6, 6, 4, 4, 5, 6, 8)

FIRs <- c(0, 0, 1, rep(0, 6),
          rep(0, 7), 1, 1)
```

```r
GIRs <- c(0, 0, 0, 1, 0, 0, 1, 0, 0,
          0, 1, 0, 0, 1, rep(0, 4))

putts_rec <- c(rep(2, 8), 3,
               1, 2, 2, 2, 2, 1, 2, 3, 2)

chips_rec <- c(1, 1, 1, 0, 0, 1, 0, 1, 3,
               1, 0, 2, 1, 0, 2, 1, 1, 4)

penalties_rec <- c(rep(0, 18))

tee_clubs <- c('D', 'D', 'D', 'D', 'D', '9', 'D', '5', 'D',
               'D', '7', 'D', 'D', 'D', '7', 'D', 'D', 'D')

index <- 11.0
```

Specify Course and Tees

```r
# get the scorecard for the new round, ensuring hole-by-hole scores are filled
if ( length(hole_scores) > 0 ) {
 Card <- golf::get_course(course = round_course, date = round_date, tees = round_tees)
}
```

Get Scoring Metrics

```r
# fill the scorecard for the new round, ensuring hole-by-hole scores are filled
if ( length(hole_scores) > 0 ) {
 Card <- golf::log_score(Scorecard = Card,
                       hole_by_hole = hole_scores,
                       GHIN = 10526424,
                       index = index,
                       FIR = FIRs,
                       GIR = GIRs,
                       putts = putts_rec,
                       chips = chips_rec,
                       penalties = penalties_rec,
                       tee_club = tee_clubs)
}
```

Update the db

Update the Players Table

```r
con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'players' table
```

```r
if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DES
     dplyr::distinct(date) |>
     unlist() %>%
     lubridate::as_date(.) |>
     as.character() < round_date &

     length(hole_scores) > 0
     ) {

  DBI::dbAppendTable(conn = con,
                   name = 'players',
                   value = players <- Card |>
                     dplyr::select(GHIN, index, date) |>
                     dplyr::distinct() |>
                     dplyr::rename(handicap_index = index) |>
                     dplyr::mutate(date = as.character(date))
                   )

}
```

Update the Courses Table

```r
con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'courses' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DES
     dplyr::distinct(date) |>
     unlist() %>%
     lubridate::as_date(.) |>
     as.character() < round_date &

     length(hole_scores) > 0
     ) {
  DBI::dbAppendTable(conn = con,
                   name = 'courses',
                   value = course <- Card |>
                     dplyr::select(course, tees, to_par, slope, course_rating, hole, yds, par, hole_hand
                     dplyr::distinct() |>
                     dplyr::rename(course_name = course)
                   )
}
```

Update the Rounds Table

```r
con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'rounds' table
```

```r
if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC
     dplyr::distinct(date) |>
     unlist() %>%
     lubridate::as_date(.) |>
     as.character() < round_date &

     length(hole_scores) > 0
     ) {
  DBI::dbAppendTable(conn = con,
                   name = 'rounds',
                   value = Card |>
                     dplyr::select(-to_par, -slope, -course_rating, -yds, -par) |>
                     dplyr::distinct() |>
                     dplyr::rename(handicap_index = index) |>
                     dplyr::rename(course_name = course) |>
                     dplyr::mutate(date = as.character(date))
                   )
}
```

Get Club Metrics df

```r
# create a dataframe, specifying the shape for tracked shots on each hole
  # -> all strokes from off the green

club_metrics_df <- golf::get_tracked_shots_data_shape(round_date = round_date) |>
  dplyr::mutate(hole = gsub(hole, pattern = 'hole_', replacement = '') |> as.numeric()) |>
  dplyr::arrange(hole) |>
  dplyr::mutate(hole = paste0("hole_", hole))
```

Annotate Club Metrics

```r
# manually annotate from Garmin Golf App log
club_choice <- c(
  'D', 'GW', 'PW',
  'D', '5', 'SW',
  'D', '3W', 'GW', 'PW',
  'D', '6',
  'D', '4', 'PW',
  '9', 'GW',
  'D', '7',
  '5', 'SW',
  'D', '3W', 'SW', 'SW', 'LW',

  'D', 'GW', 'P',
  '7',
  'D', 'GW', 'GW', 'P',
  'D', '3W', '7', 'GW',
  'D', '8',
  '7', 'PW', 'PW',
```

```
  'D', '3W', '7',
  'D', 'GW', 'GW',
  'D', '7', 'GW', '7', 'GW', 'LW'
)

dist_to_target <- c(
  275, 95, 10,
  275, 215, 25,
  275, 250, 89, 16,
  275, 182,
  275, 220, 145,
  135, 40,
  275, 92,
  200, 65,
  275, 250, 65, 45, 15,

  275, 115, 5,
  185,
  275, 120, 50, 3,
  275, 250, 75, 50,
  275, 165,
  185, 30, 8,
  275, 250, 30,
  275, 115, 55,
  275, 185, 20, 30, 25, 10
)

yds <- c(
  264, 83, 11,
  195, 222, 17,
  247, 242, 72, 19,
  240, 182,
  225, 74, 145,
  97, 35,
  263, 92,
  221, 64,
  224, 183, 125, 72, 10,

  294, 115, 5,
  186,
  276, 73, 60, 6,
  127, 281, 76, 46,
  277, 154,
  160, 38, 6,
  285, 257, 35,
  279, 62, 55,
  255, 154, 16, 18, 23, 7
)

lie_type <- c(
  'tee', 'rough', 'fairway',
  'tee', 'rough', 'rough',
  'tee', 'fairway', 'fairway', 'fairway',
```

```
  'tee', 'rough',
  'tee', 'rough', 'rough',
  'tee', 'rough',
  'tee', 'rough',
  'tee', 'fairway',
  'tee', 'rough', 'rough', 'rough', 'sand',

  'tee', 'rough', 'fairway',
  'tee',
  'tee', 'rough', 'fairway', 'fairway',
  'tee', 'rough', 'rough', 'fairway',
  'tee', 'rough',
  'tee', 'fairway', 'fairway',
  'tee', 'fairway', 'rough',
  'tee', 'fairway', 'fairway',
  'tee', 'fairway', 'rough', 'rough', 'rough', 'sand'
)

target_status <- c(
  'no', 'no', 'yes',
  'no', 'no', 'yes',
  'yes', 'yes', 'no', 'yes',
  'no', 'yes',
  'no', 'no', 'yes',
  'no', 'yes',
  'no', 'yes',
  'no', 'yes',
  'no', 'no', 'no', 'no', 'yes',

  'no', 'no', 'yes',
  'yes',
  'no', 'no', 'no', 'yes',
  'no', 'no', 'yes', 'yes',
  'no', 'yes',
  'no', 'no', 'yes',
  'yes', 'no', 'yes',
  'yes', 'no', 'yes',
  'yes', 'no', 'no', 'no', 'no', 'yes'
)

location <- c(
  'left', 'short', 'on_target',
  'left', 'right', 'on_target',
  'on_target', 'on_target', 'short', 'on_target',
  'right', 'on_target',
  'left', 'short', 'on_target',
  'right', 'on_target',
  'right', 'on_target',
  'right', 'on_target',
  'left', 'short', 'long', 'right', 'on_target',

  'left', 'right', 'on_target',
  'on_target',
```

```
  'long', 'short', 'long', 'on_target',
  'left', 'right', 'on_target', 'on_target',
  'left', 'on_target',
  'short', 'long', 'on_target',
  'on_target', 'right', 'on_target',
  'on_target', 'short', 'on_target',
  'on_target', 'right', 'right', 'short', 'short', 'on_target'
)

type_of_shot <- c(
  'tee', 'choked', 'chip',
  'tee', 'full', 'chip',
  'tee', 'full', 'choked', 'chip',
  'tee', 'full',
  'tee', 'full', 'full',
  'tee', 'chip',
  'tee', 'punch',
  'tee', 'chip',
  'tee', 'full', 'choked', 'chip', 'gsbunker',

  'tee', 'full', 'chip',
  'tee',
  'tee', 'full', 'chip', 'chip',
  'tee', 'full', 'punch', 'chip',
  'tee', 'full',
  'tee', 'chip', 'chip',
  'tee', 'full', 'chip',
  'tee', 'full', 'chip',
  'tee', 'full', 'chip', 'punch', 'chip', 'gsbunker'
)
```

Get Shot Metrics

```
##    tracked_shots sum(tracked_shots)
## 1              3                 55
## 2              3                 55
## 3              4                 55
## 4              2                 55
## 5              3                 55
## 6              2                 55
## 7              2                 55
## 8              2                 55
## 9              5                 55
## 10             3                 55
## 11             1                 55
## 12             4                 55
## 13             4                 55
## 14             2                 55
## 15             3                 55
## 16             3                 55
## 17             3                 55
## 18             6                 55
```

Update the Club Metrics Table

```
con <- golf::get_db_connection()

# if the data is newer than the last round in the 'club_metrics' table, append
  # the new data to the table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY da
     dplyr::distinct(date) |>
     unlist() %>%
     lubridate::as_date(.) |>
     as.character() < round_date &

     length(hole_scores) > 0
     ) {
  DBI::dbAppendTable(conn = con,
                   name = 'club_metrics',
                   value = club_metrics |>
                     dplyr::mutate(date = as.character(date))
                   )
}
```

```
# clean the global environment
rm(list = ls()[which(grepl(ls(), pattern= 'con|round_course|round_date|index')==F)])
```

Summarize Metrics

Gather and Format

Gather and format from the database

```
con <- golf::get_db_connection()

# get the hole-by-hole scores from each round in the database

  # join the hole-by-hole scores from the 'rounds' table to the 'courses' table
    # need each hole's par rating
    # need each course's course_rating from the 'courses' table

  # join the 'rounds' and 'courses' tables to the 'players' table
    # need handicap_index from the 'players' table

scores <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT sub.* FROM
  (SELECT DISTINCT r.*, c.par, c.course_rating FROM rounds r
  LEFT JOIN courses c
  ON c.tees = r.tees
  AND c.course_name = r.course_name
  AND c.hole = r.hole
  AND c.hole_handicap = r.hole_handicap) AS sub
  INNER JOIN players p
```

```
  ON sub.GHIN = p.GHIN
  AND sub.handicap_index = p.handicap_index
  AND sub.date = p.date;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date), # convert strings to date's
                hole = stringr::str_extract(hole, pattern = '[0-9]{1,}') |> # enforce hole order
                  as.numeric()) |>
  dplyr::relocate(par, .after = hole) |>
  dplyr::relocate(course_rating, .after = tees) |>
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole) |>
  dplyr::ungroup()
```

```
con <- golf::get_db_connection()

# get stroke metrics from the 'club_metrics' table

stroke_quality <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT * FROM club_metrics;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date), # convert strings to date's
                hole = stringr::str_extract(hole, pattern = '[0-9]{1,}') |> # enforce hole order
                  as.numeric()) |>
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole, stroke) |>
  dplyr::ungroup()
```

Compute Advanced Metrics

Compute more nuanced metrics

```
# define metrics
scores_sum <- scores |>
  dplyr::mutate(date_course = paste0(date,
                                     '\n',
                                     course_name, '\n',
                                     handicap_index),
                chips = dplyr::case_when(
                  grepl(date, pattern = '07-13') ~ NA_real_, TRUE ~ chips),
                `chips+putts` = chips+putts,
                FIR_opps = dplyr::case_when(par > 3 ~ 1, TRUE ~ NA),
                `Iron FIRs` = dplyr::case_when(par > 3 &
                                                 grepl(tee_club, pattern = '4|5') &
                                                 FIR == 1 ~ 1,
                                               TRUE ~ 0),
                `Iron FIR opps` = dplyr::case_when(par > 3 &
                                                     grepl(tee_club, pattern = '4|5') ~ 1,
                                                   TRUE ~ 0),
                `Driver FIRs` = dplyr::case_when(par > 3 &
                                                   tee_club == 'D' &
                                                   FIR == 1 ~ 1,
                                                 TRUE ~ 0),
```

```
               `Driver FIR opps` = dplyr::case_when(par > 3 &
                                                  tee_club == 'D' ~ 1,
                                                TRUE ~ 0),
               `Par 3 GIRs` = dplyr::case_when(par == 3 &
                                                 GIR == 1 ~ 1,
                                               TRUE ~ 0),
               greenie_putts = dplyr::case_when(GIR == 1 ~ putts, TRUE ~ NA_real_),
               updown_conv = dplyr::case_when(GIR == 0 &
                                                par == gross ~ 1,
                                              TRUE ~ 0),
               updown_opps = dplyr::case_when(GIR == 0 &
                                                chips > 0 ~ 1,
                                              TRUE ~ 0)
               ) |>
  dplyr::rename(`Handicap Index` = handicap_index)

# summarize by round
scores_sum <- scores_sum |>
  dplyr::group_by(date, date_course, course_rating, `Handicap Index`) |>
  dplyr::summarize(
                  FIRs = sum(FIR, na.rm = F),
                  `Iron FIRs` = sum(`Iron FIRs`, na.rm = T),
                  `Iron FIR%` = round(((sum(`Iron FIRs`, na.rm = T)/sum(`Iron FIR opps`, na.rm = T))*10
                  `Driver FIRs` = sum(`Driver FIRs`, na.rm = T),
                  `Driver FIR%` = round(((sum(`Driver FIRs`, na.rm = T)/sum(`Driver FIR opps`, na.rm =
                  `FIR%` = round((sum(FIRs, na.rm = T)/sum(FIR_opps, na.rm = T))*100, 1),
                  GIRs = sum(GIR, na.rm = F),
                  `Par 3 GIRs` = sum(`Par 3 GIRs`, na.rm = T),
                  `GIR%` = round((sum(GIRs, na.rm = T)/18)*100, 1),
                  putts = sum(putts, na.rm = F),
                  `Avg GIR putts` = round(mean(greenie_putts, na.rm = T), 2),
                  chips = sum(chips, na.rm = F),
                  `chips+putts` = sum(`chips+putts`, na.rm = F),
                  `UpDown%` = round(((sum(updown_conv, na.rm = T)/sum(updown_opps, na.rm = T))*100), 1
                  pars = sum(is_gross_par, na.rm = F),
                  birdies = sum(is_gross_birdie, na.rm = F),
                  bogies = sum(is_gross_bogey, na.rm = F),
                  `doubles+` = sum(is_gross_bogey_worse, na.rm = F),
                  penalties = sum(penalties, na.rm = T),
                  `Gross Score` = mean(tot_gross, na.rm = F),
                  `Net Score` = mean(tot_net, na.rm = F)) %>%
  dplyr::mutate(dplyr::across(c(`Iron FIRs`, `Driver FIRs`,
                                `Iron FIR%`, `Driver FIR%`, `FIR%`),
                              ~dplyr::if_else(is.na(FIRs), NA, .)),

               dplyr::across(c(`Par 3 GIRs`, `Avg GIR putts`,
                                `UpDown%`, `GIR%`),
                              ~dplyr::if_else(is.na(GIRs), NA, .)),

               `chips+putts` = dplyr::case_when(is.na(chips) ~ NA_real_,
                                                 TRUE ~ `chips+putts`)) %>%

  dplyr::mutate(
```

```
    `UpAndDown%` = dplyr::case_when(

      grepl(date, pattern = '07-13|09-21') ~ NA, TRUE ~ `UpDown%`),

    `Iron FIR%` = dplyr::case_when(`Iron FIR%` == NaN ~ 0.0, TRUE ~ `Iron FIR%`))
```

```
head(scores_sum |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 26
## # Groups:   date, date_course, course_rating [6]
##   date       date_course             course_rating `Handicap Index`  FIRs `Iron FIRs` `Iron FIR%` `Di
##   <date>     <chr>                           <dbl>            <dbl> <int>       <dbl>       <dbl>
## 1 2026-04-19 "2026-04-19\nRandolph ~          71.7             11       3           0       NaN
## 2 2026-04-05 "2026-04-05\nRandolph ~          71.7             11       6           0         0
## 3 2026-03-29 "2026-03-29\nDell Uric~          70.3             10       5           2        66.7
## 4 2026-03-08 "2026-03-08\nRandolph ~          71.7             10       2           0         0
## 5 2026-02-22 "2026-02-22\nDell Uric~          68               10.1     7           1        33.3
## 6 2026-02-08 "2026-02-08\nRandolph ~          70               10.2     2           0         0
## # i 1 more variable: `UpAndDown%` <dbl>
```

View Metrics

Separate and view metrics

Scoring Metrics Scores and Handicap

```
scoring_metrics <- scores_sum |>
  dplyr::select(`Handicap Index`, `Gross Score`, `Net Score`)
head(scoring_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 6
## # Groups:   date, date_course, course_rating [6]
##   date       date_course                      course_rating `Handicap Index` `Gross Score` `Net Sco
##   <date>     <chr>                                    <dbl>            <dbl>         <dbl>       <d
## 1 2026-04-19 "2026-04-19\nRandolph North\n11"          71.7             11              91
## 2 2026-04-05 "2026-04-05\nRandolph North\n11"          71.7             11              88
## 3 2026-03-29 "2026-03-29\nDell Urich\n10"              70.3             10              82
## 4 2026-03-08 "2026-03-08\nRandolph North\n10"          71.7             10              84
## 5 2026-02-22 "2026-02-22\nDell Urich\n10.1"            68               10.1            92
## 6 2026-02-08 "2026-02-08\nRandolph North\n10.2"        70               10.2            83
```

Stroke Metrics Pars, birdies, bogies, etc.

```
stroke_metrics <- scores_sum |>
  dplyr::select(`doubles+`, bogies, pars, birdies)
head(stroke_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 7
## # Groups:   date, date_course, course_rating [6]
##   date       date_course                          course_rating `doubles+` bogies  pars birdies
##   <date>     <chr>                                        <dbl>      <int>  <int> <int>   <int>
## 1 2026-04-19 "2026-04-19\nRandolph North\n11"              71.7          4      8     6       0
## 2 2026-04-05 "2026-04-05\nRandolph North\n11"              71.7          3      9     6       0
## 3 2026-03-29 "2026-03-29\nDell Urich\n10"                  70.3          4      5     7       2
## 4 2026-03-08 "2026-03-08\nRandolph North\n10"              71.7          3      6     9       0
## 5 2026-02-22 "2026-02-22\nDell Urich\n10.1"                68            6      7     5       0
## 6 2026-02-08 "2026-02-08\nRandolph North\n10.2"            70            2      8     7       1
```

Around-the-Green Metrics Chips, putts, etc.

```
atg_metrics <- scores_sum |>
  dplyr::select(chips, `chips+putts`, `UpAndDown%`, putts, `Avg GIR putts`)
head(atg_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 8
## # Groups:   date, date_course, course_rating [6]
##   date       date_course                          course_rating chips `chips+putts` `UpAndDown%` putts
##   <date>     <chr>                                        <dbl> <dbl>         <dbl>        <dbl> <int>
## 1 2026-04-19 "2026-04-19\nRandolph North\n11"              71.7    20            56         15.4    36
## 2 2026-04-05 "2026-04-05\nRandolph North\n11"              71.7    21            52         20      31
## 3 2026-03-29 "2026-03-29\nDell Urich\n10"                  70.3    15            46         20      31
## 4 2026-03-08 "2026-03-08\nRandolph North\n10"              71.7    12            45         30      33
## 5 2026-02-22 "2026-02-22\nDell Urich\n10.1"                68      18            52         16.7    34
## 6 2026-02-08 "2026-02-08\nRandolph North\n10.2"            70      11            46         22.2    35
```

Ball Striking Metrics Approach and tee accuracy

```
ball_striking_metrics <- scores_sum |>
  dplyr::select(GIRs, `GIR%`, `Par 3 GIRs`,
                FIRs, `FIR%`, `Iron FIRs`, `Iron FIR%`,
                `Driver FIRs`, `Driver FIR%`)
head(ball_striking_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 12
## # Groups:   date, date_course, course_rating [6]
##   date       date_course                          course_rating  GIRs `GIR%` `Par 3 GIRs`  FIRs `FIR%`
##   <date>     <chr>                                        <dbl> <int>  <dbl>        <dbl> <int>  <dbl>
## 1 2026-04-19 "2026-04-19\nRandolph North\n11"              71.7     4   22.2            1     3   21.4
## 2 2026-04-05 "2026-04-05\nRandolph North\n11"              71.7     3   16.7            1     6   42.9
## 3 2026-03-29 "2026-03-29\nDell Urich\n10"                  70.3     7   38.9            1     5   38.5
## 4 2026-03-08 "2026-03-08\nRandolph North\n10"              71.7     7   38.9            2     2   14.3
## 5 2026-02-22 "2026-02-22\nDell Urich\n10.1"                68       3   16.7            1     7   53.8
## 6 2026-02-08 "2026-02-08\nRandolph North\n10.2"            70       7   38.9            3     2   14.3
```

Club Metrics Yardage and accuracy for each club

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date       club      n rd_min_yds_to_target rd_min_yds_traveled rd_min_yd_diff
##   <date>     <chr> <int>                <dbl>               <dbl>          <dbl>
## 1 2026-04-19 3W        4                  250                 183            -31
## 2 2026-04-19 4         1                  220                  74            146
## 3 2026-04-19 5         2                  200                 221            -21
## 4 2026-04-19 6         1                  182                 182              0
## 5 2026-04-19 7         7                   30                  18             -5
## 6 2026-04-19 8         1                  165                 154             11
```

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date       club      n rd_max_yds_to_target rd_max_yds_traveled rd_max_yd_diff
##   <date>     <chr> <int>                <dbl>               <dbl>          <dbl>
## 1 2026-04-19 3W        4                  250                 281             67
## 2 2026-04-19 4         1                  220                  74            146
## 3 2026-04-19 5         2                  215                 222             -7
## 4 2026-04-19 6         1                  182                 182              0
## 5 2026-04-19 7         7                  185                 186             31
## 6 2026-04-19 8         1                  165                 154             11
```

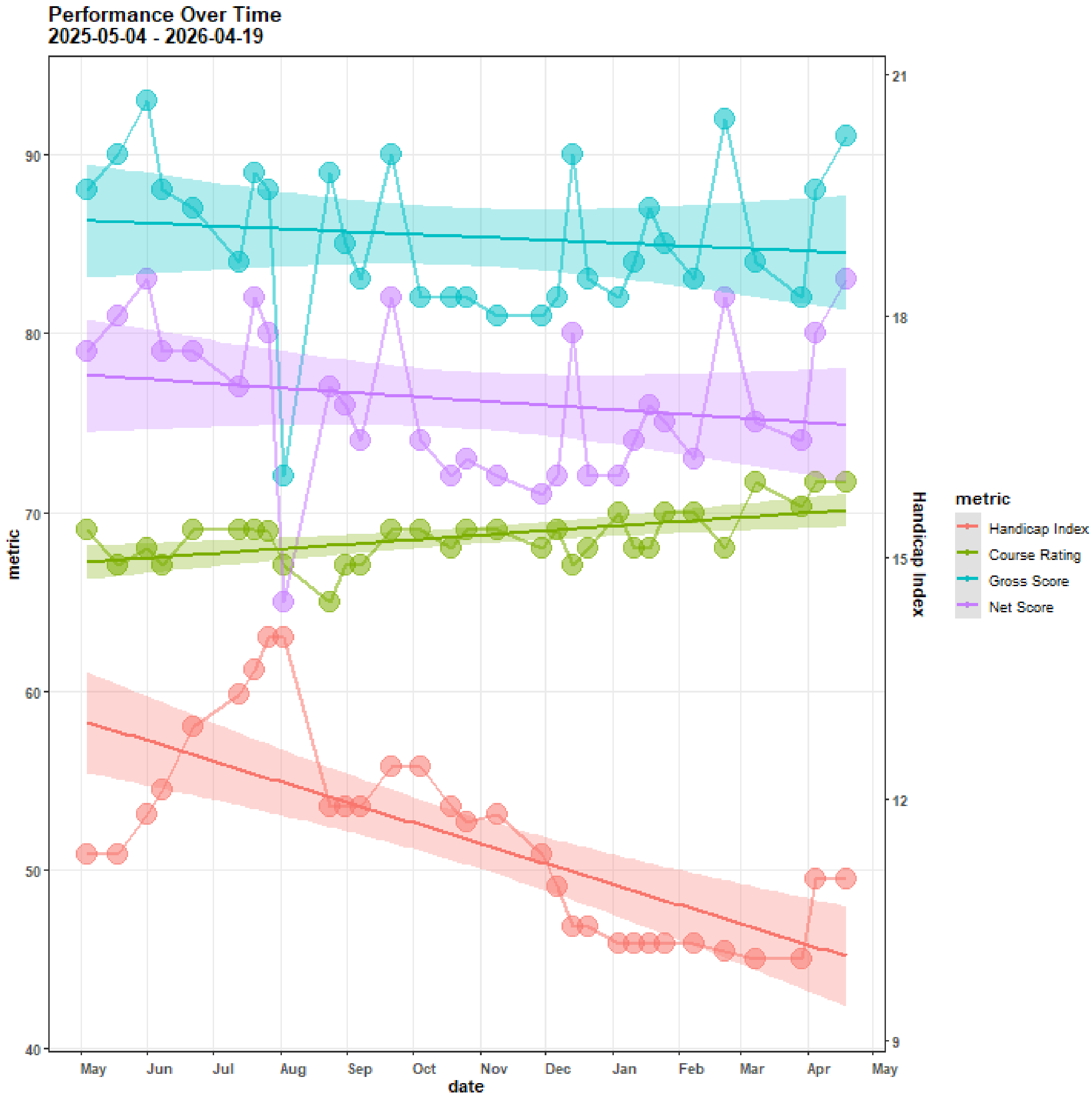
```
## # A tibble: 6 x 10
## # Groups:   date [1]
##   date       club  `rd club strokes` miss_direction `rd club miss dir` rd_avg_yds_to_target rd_avg_y
##   <date>     <chr>             <int> <chr>                       <int>                <dbl>
## 1 2026-04-19 3W                    4 on_target                       1                  250
## 2 2026-04-19 3W                    4 right                           2                  250
## 3 2026-04-19 3W                    4 short                           1                  250
## 4 2026-04-19 4                     1 short                           1                  220
## 5 2026-04-19 5                     2 right                           2                  208.
## 6 2026-04-19 6                     1 on_target                       1                  182
```

Plot Metric Summaries

Scoring Metrics

```
scoring_metrics |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
                `Handicap Index` = `Handicap Index`*4.5) |>
  dplyr::rename(`Course Rating` = course_rating) |>
  dplyr::group_by(date, date_course, `Handicap Index`) |>
  tidyr::pivot_longer(cols = c(`Course Rating`:`Net Score`), names_to = 'metric', values_to = 'value', v
  ggplot2::ggplot(aes(x = date_course, y = value)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1,
                        color = factor(metric,
                                       levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'N
                        fill = factor(metric,
                                       levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'N
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1,
                        color = factor(metric,
```

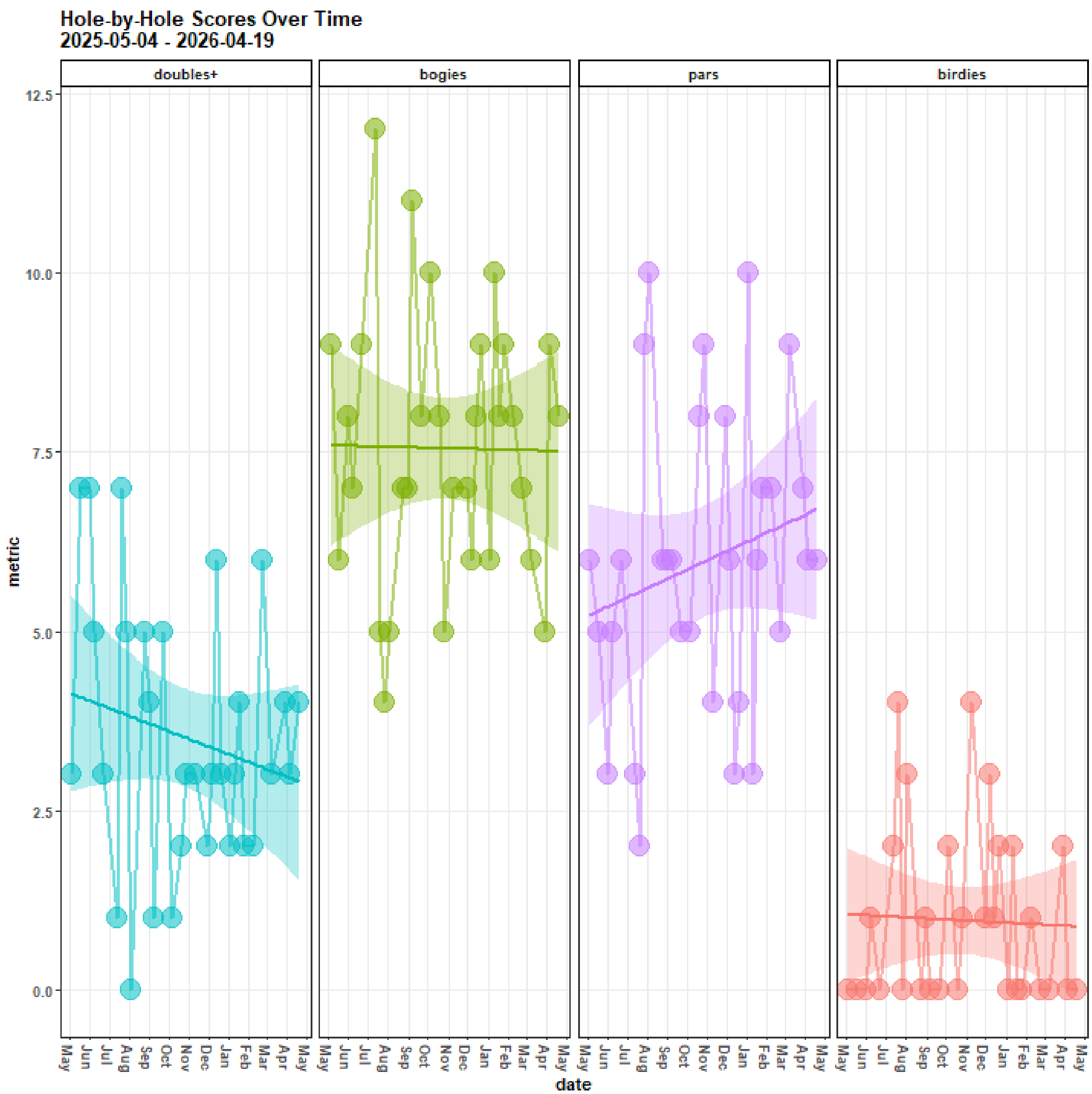
```
                                            levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'N
                            fill = factor(metric,
                                            levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'N
  ggplot2::geom_smooth(aes(x = date, y = value, group = metric,
                             color = factor(metric,
                                             levels = c('Handicap Index', 'Course Rating', 'Gross Score',
                             fill = factor(metric,
                                            levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'N
                         alpha = 0.3, method = 'lm') +
  ggplot2::scale_y_continuous(sec.axis = ggplot2::sec_axis(~./4.5, name = 'Handicap Index'))+
  ggplot2::labs(title = paste0('Performance Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
                                 unlist() %>% lubridate::as_date(.) |>
                                 as.character(),
                               ' - ',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::first() |>
                                 unlist() %>% lubridate::as_date(.) |>
                                 as.character()),
                x = 'date',
                y = 'metric',
                color = 'metric'
                ) +
  ggplot2::guides(alpha = 'none', size = 'none', fill = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(panel.grid.minor = ggplot2::element_blank(),
                 axis.title = ggplot2::element_text(face = 'bold'),
                 axis.text.y = ggplot2::element_text(face = 'bold'),
                 axis.text.x = ggplot2::element_text(face = 'bold', angle = 0, hjust = 0, vjust = 0),
                 title = ggplot2::element_text(face = 'bold')
                 ) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b')
```



Stroke Metrics

```
stroke_metrics |>
  tidyr::pivot_longer(cols = c(`doubles+`:birdies), names_to = 'metric', values_to = 'value', values_dro
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Hole-by-Hole Scores Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
```

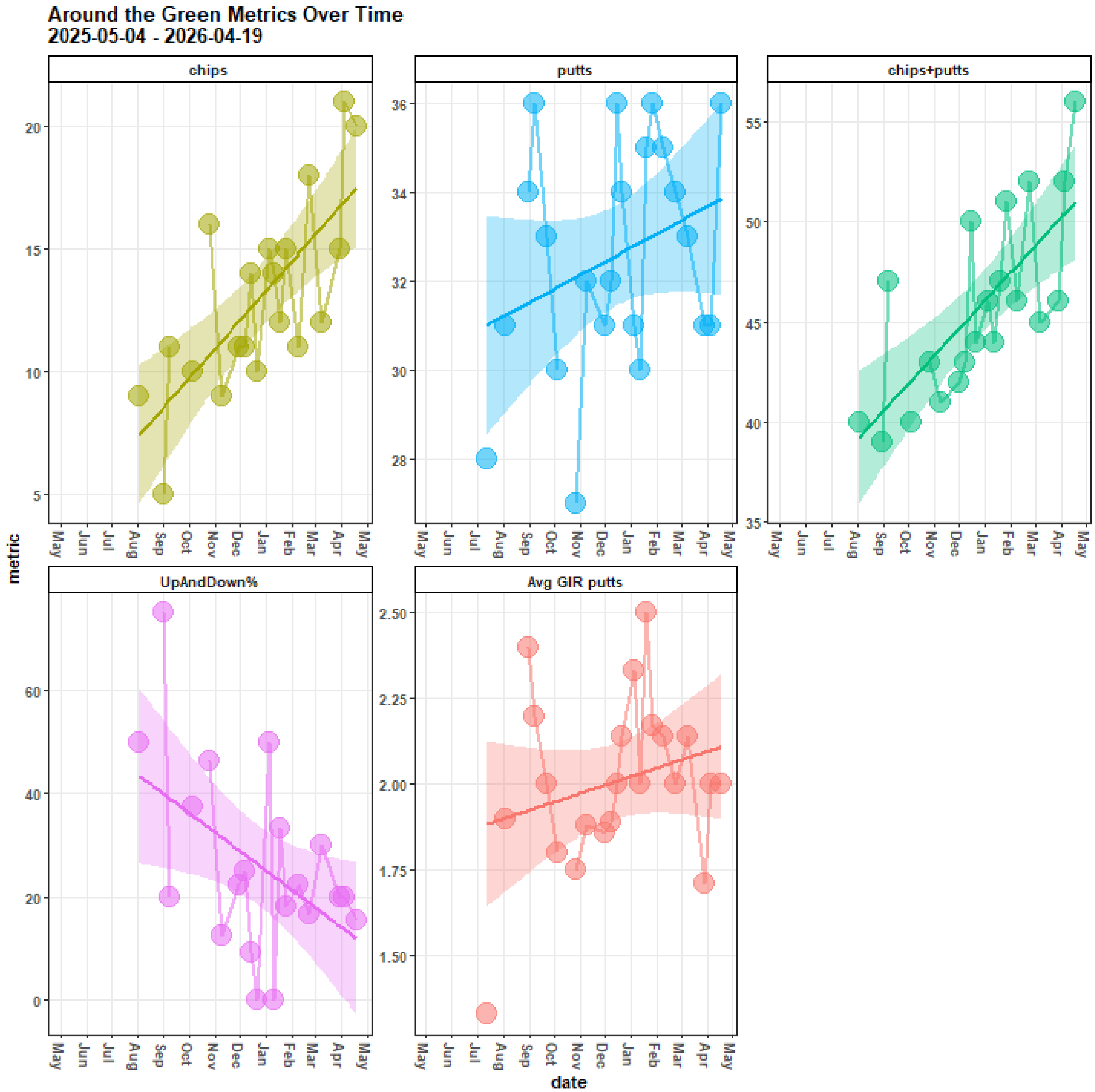
```
                                 unlist() %>% lubridate::as_date(.) |>
                                 as.character(),
                               ' - ',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::first() |>
                                 unlist() %>% lubridate::as_date(.) |>
                                 as.character()),
              x = 'date',
              y = 'metric',
              color = 'metric', fill = 'metric') +
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                        fill = 'white')) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
  ggplot2::facet_wrap(~factor(metric, levels = c('doubles+', 'bogies', 'pars', 'birdies')), nrow = 1)
```



Around the Green Metrics

```
atg_metrics |>
  tidyr::pivot_longer(cols = c(chips:`Avg GIR putts`), names_to = 'metric', values_to = 'value', values_
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Around the Green Metrics Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
```

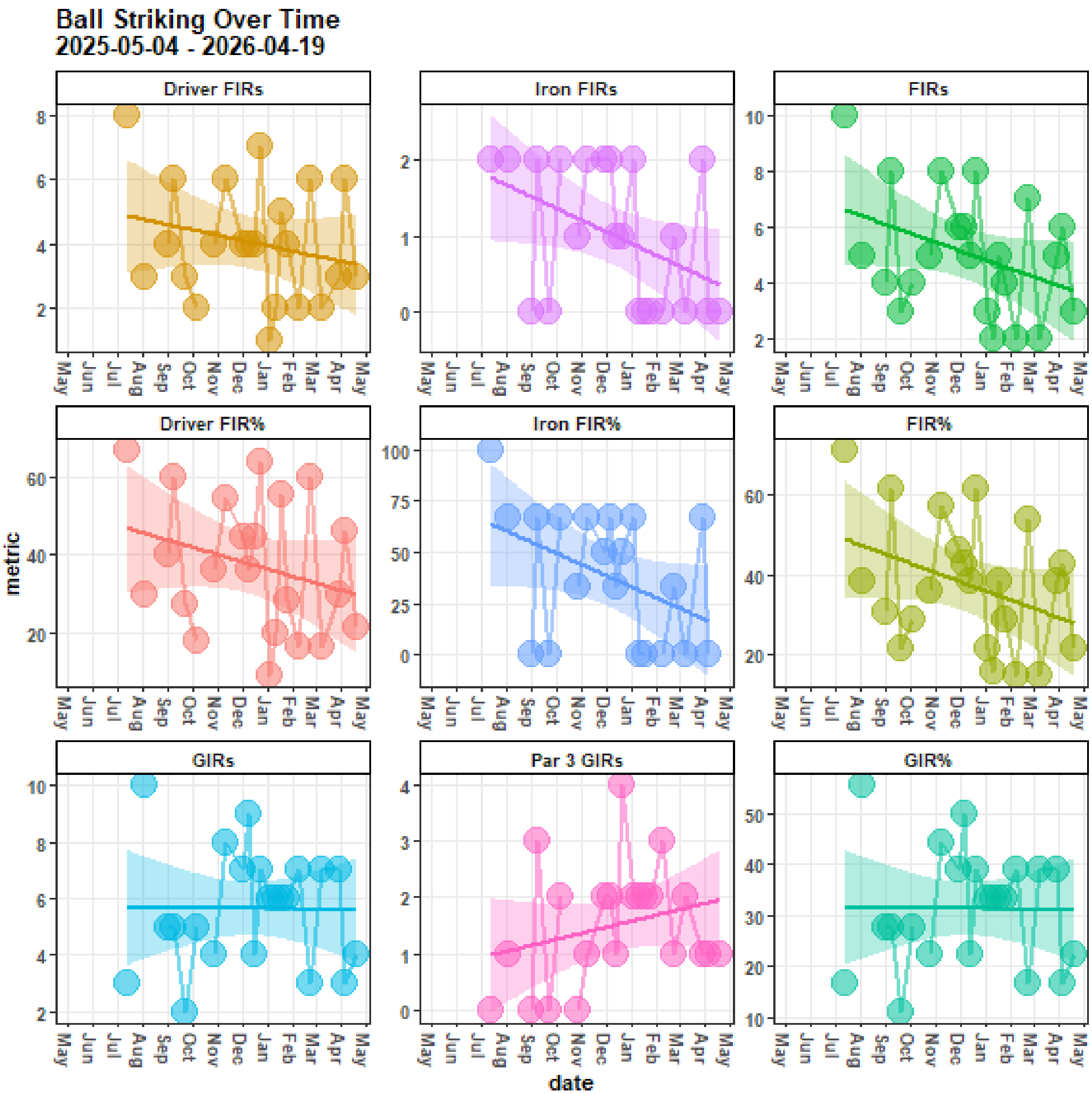
```
                                 unlist() %>% lubridate::as_date(.) |>
                                 as.character(),
                               ' - ',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::first() |>
                                 unlist() %>% lubridate::as_date(.) |>
                                 as.character()),
              x = 'date',
              y = 'metric',
              color = 'metric', fill = 'metric') +
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                         fill = 'white')) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
  ggplot2::facet_wrap(~factor(metric, levels = c('chips', 'putts', 'chips+putts', 'UpAndDown%', 'Avg GI
```



Ball Striking Metrics

```
ball_striking_metrics |>
  tidyr::pivot_longer(cols = c(GIRs:`Driver FIR%`), names_to = 'metric', values_to = 'value', values_dr
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Ball Striking Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
```

```
                                unlist() %>% lubridate::as_date(.) |>
                                as.character(),
                              ' - ',
                              scores |>
                                dplyr::distinct(date) |>
                                dplyr::first() |>
                                unlist() %>% lubridate::as_date(.) |>
                                as.character()),
                x = 'date',
                y = 'metric',
                color = 'metric', fill = 'metric') +
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(legend.position = 'none',
                 panel.grid.minor = ggplot2::element_blank(),
                 axis.title = ggplot2::element_text(face = 'bold'),
                 axis.text.y = ggplot2::element_text(face = 'bold'),
                 axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5)
                 title = ggplot2::element_text(face = 'bold'),
                 strip.text.x.top = ggplot2::element_text(face = 'bold'),
                 strip.background = ggplot2::element_rect(color = 'black',
                                                          fill = 'white')) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
  ggplot2::facet_wrap(~factor(metric,levels = c('Driver FIRs', 'Iron FIRs', 'FIRs',
                                              'Driver FIR%', 'Iron FIR%', 'FIR%',
                                              'GIRs', 'Par 3 GIRs', 'GIR%')), scales = 'free')
```



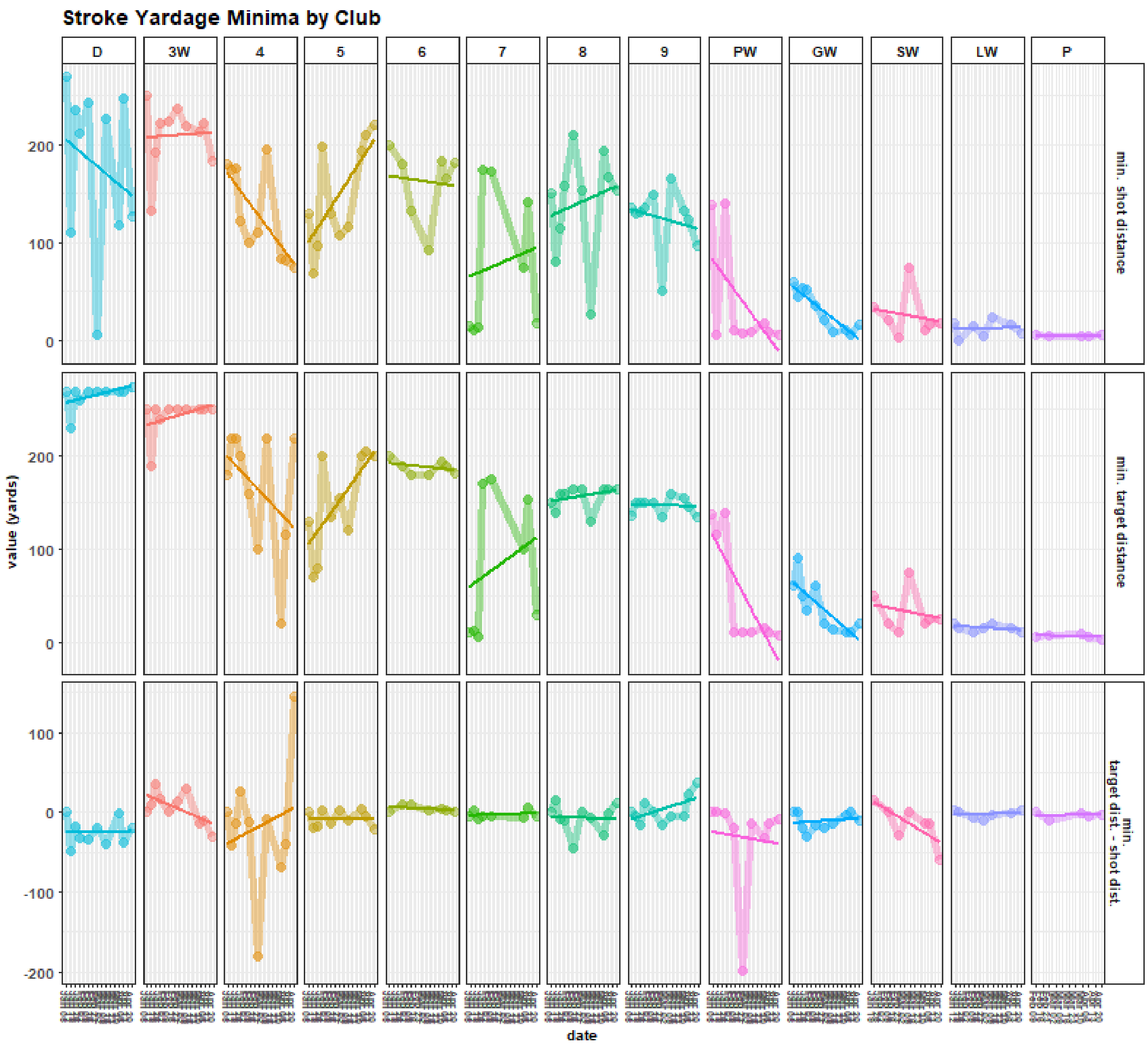
Stroke Quality Metrics

```
stroke_quality_min |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'min. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'min. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'min.\ntarget dist. - shot d
  dplyr::mutate(date = lubridate::as_date(date),
                club = as.character(club),
```

```
              n = as.integer(n)) |>
ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
ggplot2::geom_point(alpha = 0.4, size = 3) +
ggplot2::geom_line(alpha = 0.4, size = 2) +
ggplot2::geom_smooth(method = 'lm', se = F) +
ggplot2::labs(title = 'Stroke Yardage Minima by Club', x = 'date', y = 'value (yards)', fill = 'club')
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, v
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                      '6', '7', '8', '9',
                                      'PW', 'GW', 'SW', 'LW', 'P'))),
                    scales = 'free')
```



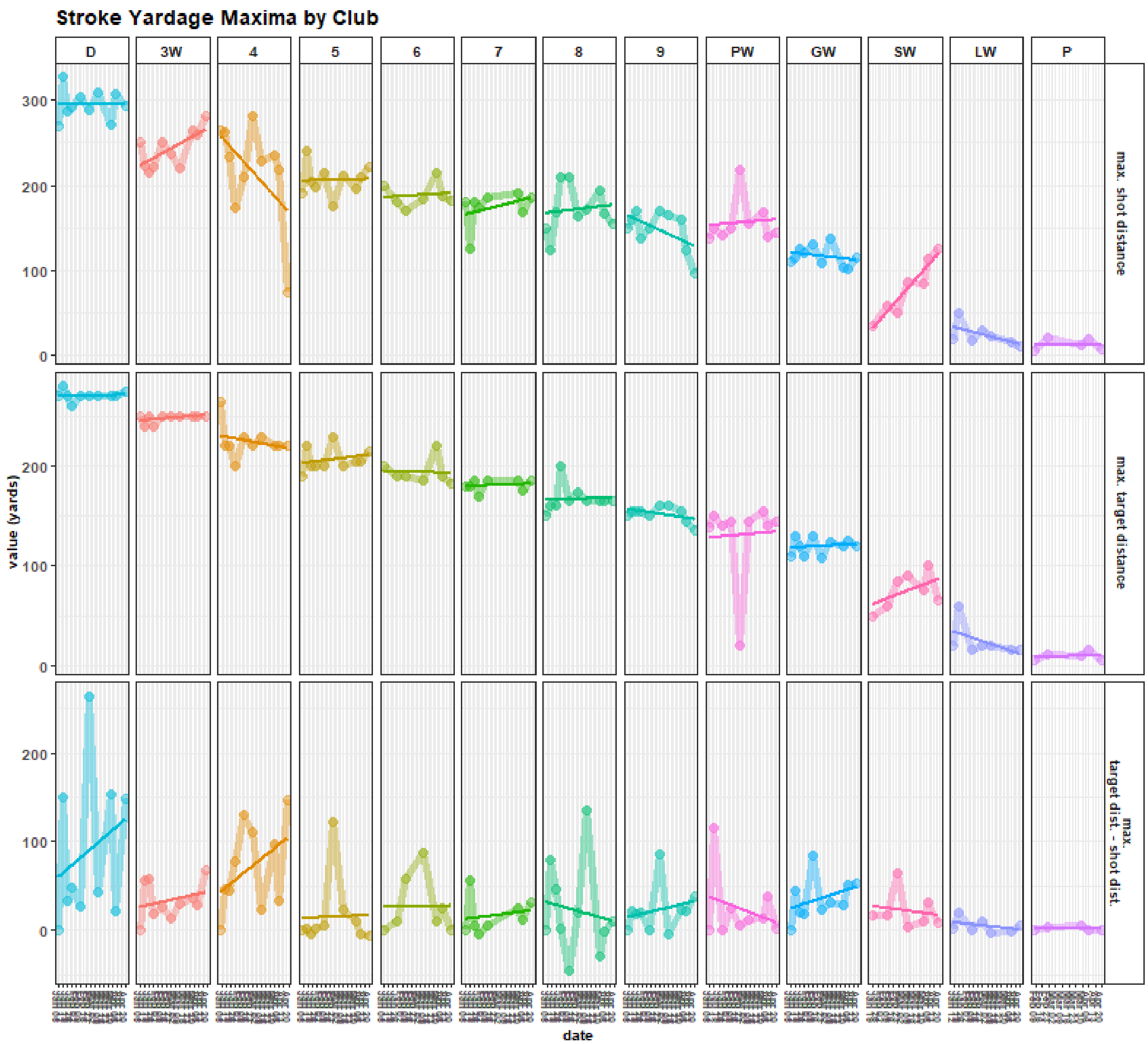
Minima

```
stroke_quality_max |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'max. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'max. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'max.\ntarget dist. - shot d
  dplyr::mutate(date = lubridate::as_date(date),
                club = as.character(club),
                n = as.integer(n)) |>
  ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
  ggplot2::geom_point(alpha = 0.4, size = 3) +
  ggplot2::geom_line(alpha = 0.4, size = 2) +
  ggplot2::geom_smooth(method = 'lm', se = F) +
  ggplot2::labs(title = 'Stroke Yardage Maxima by Club', x = 'date', y = 'value (yards)', fill = 'club')
```

```
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, v
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                      '6', '7', '8', '9',
                                      'PW', 'GW', 'SW', 'LW', 'P'))),
                    scales = 'free')
```



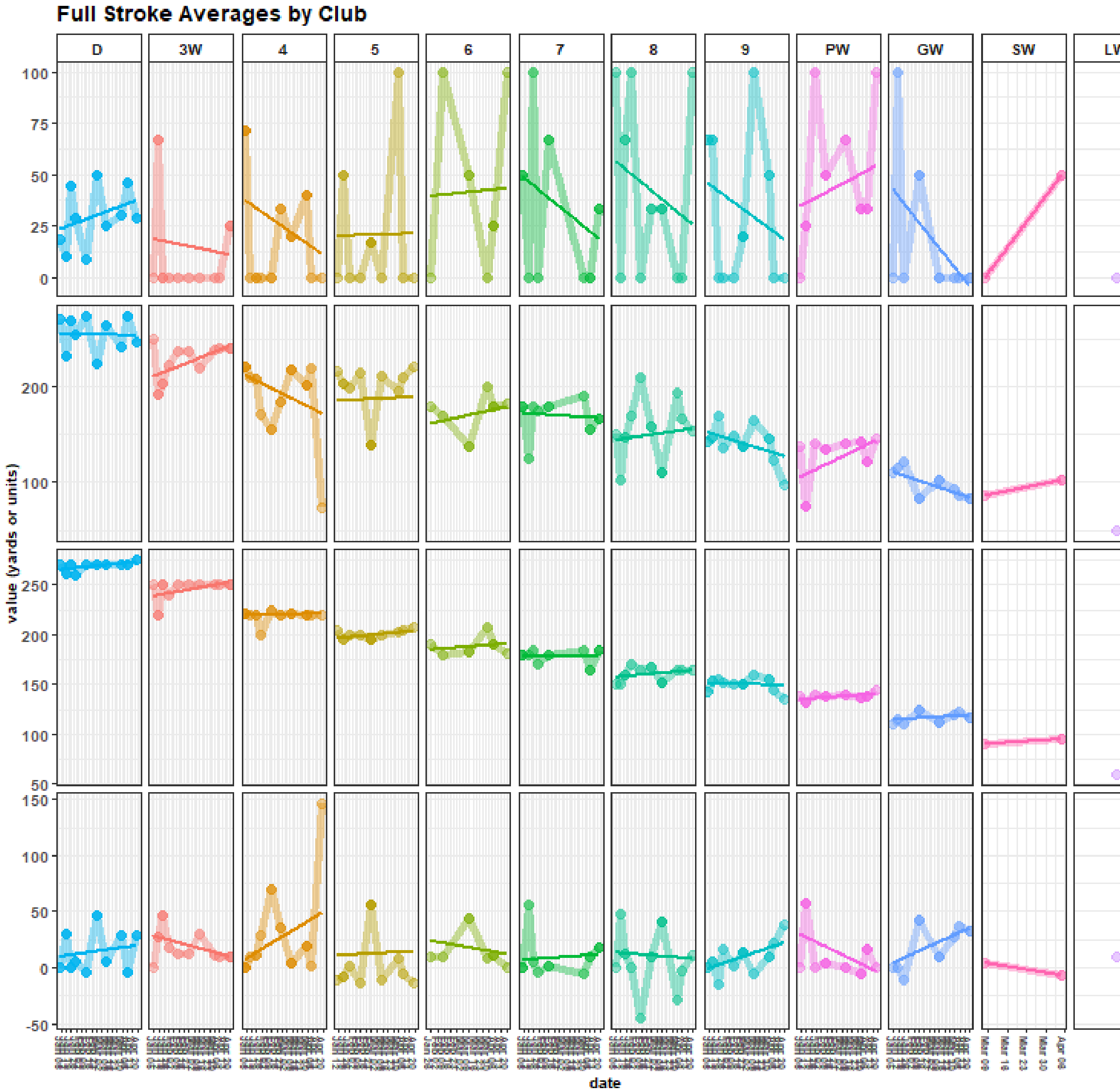
Maxima

```
full_stroke_quality_avg |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(dplyr::everything(), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains("rd")), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'avg. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'avg. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'avg.\ntarget dist. - shot d
                                          # grepl(metric, pattern = 'accuracy') ~ 'avg\naccuracy',
                                          grepl(metric, pattern = 'strokes') ~ 'n strokes',
                                          grepl(metric, pattern = 'dir') ~ 'strokes w/club\nin round\nby
                                          grepl(metric, pattern = 'acc') ~ '% accuracy')) |>
  dplyr::filter(!grepl(metric, pattern = 'strokes')) |>
  dplyr::mutate(date = lubridate::as_date(date),
                club = as.character(club),
                vals = as.numeric(vals)
```

```
              ) |>
ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
ggplot2::geom_point(alpha = 0.4, size = 3) +
ggplot2::geom_line(alpha = 0.4, size = 2) +
ggplot2::geom_smooth(method = 'lm', se = F) +
ggplot2::labs(title = 'Full Stroke Averages by Club', x = 'date', y = 'value (yards or units)', fill =
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, v
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                      '6', '7', '8', '9',
                                      'PW', 'GW', 'SW', 'LW', 'P'))),
                    scales = 'free')
```

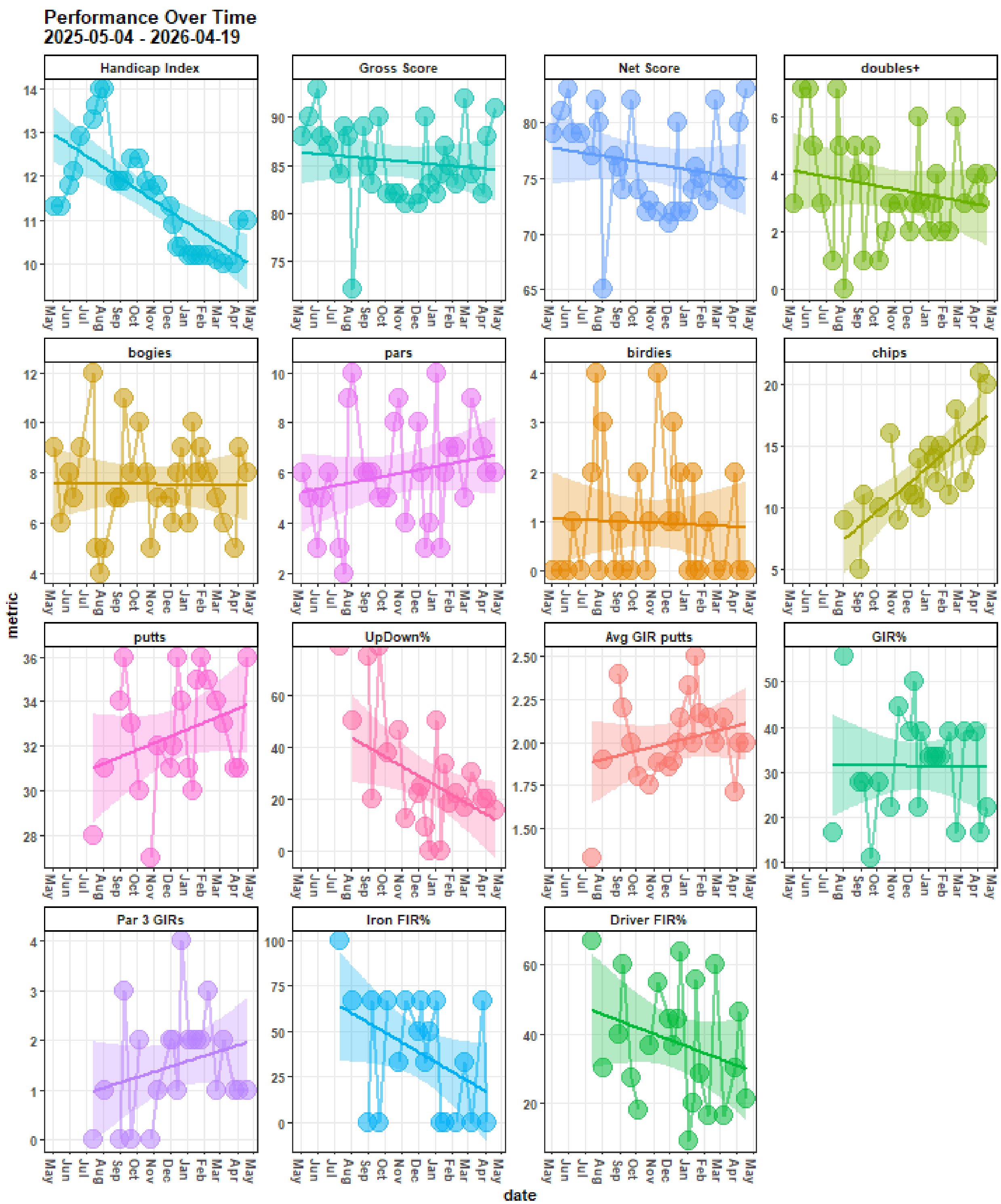


Full Stroke Average

Main Metrics

```
scores_sum |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
  tidyr::pivot_longer(cols = c(`Handicap Index`, `Gross Score`, `Net Score`,
                               `doubles+`, bogies, pars, birdies, chips, putts,
                               `UpDown%`,`Avg GIR putts`, `GIR%`, `Par 3 GIRs`,
                               `Iron FIR%`, `Driver FIR%`), names_to = 'metric', values_to = 'value', va
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Performance Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
```

```
                                  unlist() %>% lubridate::as_date(.) |>
                                  as.character(),
                                ' - ',
                                scores |>
                                  dplyr::distinct(date) |>
                                  dplyr::first() |>
                                  unlist() %>% lubridate::as_date(.) |>
                                  as.character()),
                 x = 'date',
                 y = 'metric',
                 color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                        fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric,
                    levels = c('Handicap Index', 'Gross Score', 'Net Score',
                               'doubles+', 'bogies', 'pars', 'birdies',
                               'chips', 'putts', 'UpDown%', 'Avg GIR putts',
                               'GIR%', 'Par 3 GIRs', 'Iron FIR%', 'Driver FIR%')),
                    scales = 'free',
                    ncol = 4)
```



Performance Over Time
2025-05-04 - 2026-04-19
Handicap Index
Gross Score
Net Score
doubles+
bogies
pars
birdies
chips
putts
UpDown%
Avg GIR putts
GIR%
Par 3 GIRs
Iron FIR%
Driver FIR%
metric
date